

TMR4240 Marine Control Systems I

Project Part 1: Design of a Dynamic Positioning System

Group 7

Department of Marine Technology, NTNU

[TODO: add all group member names]

Abstract—[TODO: Write the abstract last, once the results and conclusion exist. One paragraph: what was built (3-DOF DP system for R/V Gunnerus in Python), what was tested (station keeping, rotating current, setpoint change, four-corner test), and the headline result.]

[TODO: Insert your own version of the NED/body frame figure here (Fig. 1 in the assignment PDF is a template for the idea only – the report requires an original figure).]

I. INTRODUCTION

Dynamic positioning (DP) systems automatically maintain a vessel's position and heading using active thrusters rather than mooring lines or anchors, compensating for environmental disturbances such as wind, current, and waves [1]. This report documents the design, implementation, and validation of a basic three-degree-of-freedom (3-DOF) DP system for the research vessel R/V Gunnerus, developed as Project Part 1 of TMR4240 Marine Control Systems I.

The implemented system consists of five interconnected subsystems: a current load model, a wind load model, a DP controller, a reference model, and a thrust allocation algorithm. These subsystems interface with a supplied three-DOF process plant model of R/V Gunnerus and a simulation engine that handles state integration, actuator limits, and logging.

[TODO: One or two sentences on the report structure / roadmap, e.g. "Section II introduces notation and coordinate frames... Section VI presents the four mandatory simulations...". Adjust section numbers once the whole report is assembled.]

II. COORDINATE FRAMES AND NOTATION

A. Coordinate Frames

Two coordinate frames are used to describe the vessel motion. The Earth-fixed North-East-Down (NED) frame, denoted $\{n\}$, is fixed to the Earth with axes x_n pointing North, y_n pointing East, and z_n pointing Down. This frame describes the vessel's position and heading relative to its surroundings.

The body-fixed frame, denoted $\{b\}$, is attached to the vessel, with its origin at the vessel's centre of gravity. The x_b axis points forward toward the bow (surge), the y_b axis points to starboard (sway), and the z_b axis points downward.

Both frames are right-handed. With the NED convention, a positive rotation about the downward z axis is clockwise when viewed from above; consequently, the heading angle ψ is measured clockwise from North to the vessel's bow direction.

B. Six-DOF Notation and Three-DOF Reduction

The standard six-DOF marine craft state vectors are

$$\eta_6 = [x, y, z, \phi, \theta, \psi]^\top, \quad (1)$$

$$\nu_6 = [u, v, w, p, q, r]^\top, \quad (2)$$

$$\tau_6 = [X, Y, Z, K, M, N]^\top. \quad (3)$$

Table I defines every component of these vectors and the frame in which each is expressed.

TABLE I
SIX-DOF NOTATION

Symbol	Description	Frame
x, y, z	Position (North, East, Down)	NED
ϕ, θ, ψ	Roll, pitch, yaw (heading)	NED
u, v, w	Surge, sway, heave velocity	Body
p, q, r	Roll, pitch, yaw rate	Body
X, Y, Z	Surge, sway, heave force	Body
K, M, N	Roll, pitch, yaw moment	Body

This project uses a horizontal-plane three-DOF model. Heave, roll, and pitch are neglected; only surge, sway, and yaw are retained:

$$\eta = [N, E, \psi]^\top, \quad \nu = [u, v, r]^\top, \quad \tau = [X, Y, N_z]^\top, \quad (4)$$

where η is expressed in the NED frame and ν, τ are expressed in the body frame. This distinction between frames is essential throughout the report: the vessel dynamics compute accelerations in body axes, while position is integrated in the NED frame.

C. Kinematic Equation

The kinematic equation converts body-frame velocity to NED position and heading rates. It is a purely geometric transformation, not a force model. For the three-DOF case it reduces to a yaw-only rotation:

$$\dot{\eta} = J(\psi) \nu, \quad J(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (5)$$

In scalar form,

$$\dot{N} = u \cos \psi - v \sin \psi, \quad \dot{E} = u \sin \psi + v \cos \psi, \quad \dot{\psi} = r. \quad (6)$$

For this three-DOF rotation, $J^\top(\psi) = J^{-1}(\psi)$, so the transpose can be used to convert NED-frame vectors (e.g. a position error) into the body frame (e.g. a force direction). Heading errors must always be wrapped to $(-\pi, \pi]$ using

$$\text{wrap}(\alpha) = \text{atan2}(\sin \alpha, \cos \alpha). \quad (7)$$

III. CURRENT MODEL

The ocean current is defined in the NED frame by a speed V_c and a direction, using only the North and East components (the vertical component is always zero). **[TODO: State explicitly whether your implementation's direction input means the current is coming "from" that direction or flowing "towards" it, and give the resulting NED vector convention.]**

To act on the vessel, the current is transformed from the NED frame to the body frame and subtracted from the vessel velocity to obtain the relative velocity used by the plant model:

$$\nu_c^n = [V_{c,N}, V_{c,E}, 0]^\top, \quad \nu_c^b = J^\top(\psi) \nu_c^n, \quad \nu_r = \nu - \nu_c^b. \quad (8)$$

The current does not enter the vessel model as a separate generalised force; instead it acts entirely through the relative velocity ν_r inside the hydrodynamic damping terms. This means the current model's only responsibility is to produce a correct NED velocity vector $[V_{c,N}(t), V_{c,E}(t)]$ at each time step.

[TODO: Describe your current implementation here: constant vs. time-varying profile, the parameters used in Simulations 1 and 2 (speed, direction/rotation profile, duration), and justify them with reference to the mandatory test definitions.]

IV. WIND MODEL

The wind model returns a six-component generalised body-frame load vector, of which only the surge, sway, and yaw entries ($\tau_{\text{wind}}[0, 1, 5]$) affect the three-DOF plant. The wind load is computed from the *relative* wind speed and angle using a tabulated coefficient vector:

$$\tau_{\text{wind}} = U_{rw}^2 C_w(\alpha_{rw}), \quad (9)$$

where $C_w(\alpha_{rw}) \in \mathbb{R}^6$ is interpolated from the supplied wind coefficient table (generated with the method of Blendermann [2] for R/V Gunnerus; only C_x , C_y , C_ψ are non-zero for the horizontal-plane model). The relative wind velocity is the ambient wind minus the vessel velocity, expressed in the body frame:

$$\begin{bmatrix} V_{rw,x}^b \\ V_{rw,y}^b \end{bmatrix} = J^\top(\psi) \begin{bmatrix} V_{w,N} \\ V_{w,E} \end{bmatrix} - \begin{bmatrix} u \\ v \end{bmatrix}, \quad (10)$$

$$U_{rw} = \sqrt{(V_{rw,x}^b)^2 + (V_{rw,y}^b)^2}, \quad \alpha_{rw} = \text{atan2}(V_{rw,y}^b, V_{rw,x}^b). \quad (11)$$

For Project Part 1, the wind speed has a constant mean component and a slowly varying stochastic component, while the wind direction is kept constant. **[TODO: State the mean**

wind speed/direction and the slowly varying component's parameters (e.g. standard deviation, time constant, random seed) used in Simulation 1b, and justify the chosen values. Reference the wind coefficient figure from the appendix (generated in notebooks/wind_coefficients.ipynb).]

V. THRUSTER DYNAMICS

Thrust is physically produced by rotating propellers, whose open-water characteristics relate shaft speed, propeller geometry, and advance ratio to thrust and torque. For DP design purposes, this level of detail is not modelled; instead, the allocated commands (u_i, α_i) for each thruster i are mapped directly to a body wrench through the configuration matrix:

$$\tau_{\text{thr}} = B(\alpha) u. \quad (12)$$

In Part 1, the thruster dynamics block is *disabled*: the simulator does not apply thrust-rate limits, azimuth-rate limits, or automatic saturation to the allocator's output. Commands are applied exactly as computed. The static maximum thrust u_{max} for each thruster (Table II) must therefore be respected by the thrust allocator itself, not by the simulator.

TABLE II
THRUSTER CONFIGURATION (POSITIONS RELATIVE TO CG, BODY FRAME;
 $x > 0$ FORWARD, $y > 0$ STARBOARD)

Thruster	Type	x [m]	y [m]	α_0	u_{max} [kN]
Tunnel_Bow	tunnel	+12.0	0.0	90° (fixed)	32
Azimuth_1	azimuth	-13.0	+3.0	0° (free)	80
Azimuth_2	azimuth	-13.0	-3.0	0° (free)	80

[TODO: State whether you kept the thruster dynamics disabled (as required for the mandatory Part 1 tests) or additionally enabled the full actuator dynamics as an optional extension, and justify your choice.]

VI. THRUST ALLOCATION

Thrust allocation is the interface between the DP controller, which computes a desired generalised body-frame wrench, and the propulsion system, which can only produce forces at fixed physical locations and (for two of the three thrusters) freely chosen directions.

A. The Allocation Problem

The vessel carries three thrusters (Table II): one fixed-direction tunnel thruster, contributing a single degree of freedom (signed magnitude u_T), and two azimuth thrusters, each contributing two degrees of freedom (magnitude u_j and angle α_j). This gives $1 + 2 + 2 = 5$ independent actuator degrees of freedom to produce a three-component wrench $\tau_c = [F_x, F_y, M_z]^\top$, i.e. the system is over-actuated with two redundant degrees of freedom: infinitely many actuator combinations produce the same net wrench.

Solving directly for (u_j, α_j) is a nonlinear problem, because each thruster's force components involve the product of a magnitude and trigonometric functions of an unknown angle. Section VI-C removes this nonlinearity through a change of variables.

B. Configuration Matrix

For a thruster at position (x_i, y_i) producing signed thrust u_i at angle α_i (measured clockwise from x_b , per the body-frame convention of Section II), the wrench it contributes is

$$B_i(\alpha_i) = \begin{bmatrix} \cos \alpha_i \\ \sin \alpha_i \\ x_i \sin \alpha_i - y_i \cos \alpha_i \end{bmatrix}, \quad \tau = B(\alpha) u. \quad (13)$$

Building $B(\alpha)$ column by column from Table II gives

$$B(\alpha) = \begin{bmatrix} 0 & \cos \alpha_1 & \cos \alpha_2 \\ 1 & \sin \alpha_1 & \sin \alpha_2 \\ 12 & -13 \sin \alpha_1 - 3 \cos \alpha_1 & -13 \sin \alpha_2 + 3 \cos \alpha_2 \end{bmatrix} \quad (14)$$

The tunnel column, $B_{\text{Tunnel}} = [0, 1, 12]^\top$, is worth reading in words: because the tunnel thruster's angle is fixed at $\alpha_0 = 90^\circ$ (pointing along y_b), it can only ever produce a pure sway force – never surge – with its sign set by the sign of u_T . Because it sits 12m forward of the CG, that sway force acts on a comparatively long moment arm, making the tunnel a disproportionately effective yaw actuator despite its low $u_{\max} = 32$ kN. The two azimuth columns depend on the unknown angles α_1, α_2 , which is precisely the source of the problem's nonlinearity.

C. Allocation Variables: Cartesian Linearisation

Introducing Cartesian force components for each azimuth thruster,

$$F_{x,j} = u_j \cos \alpha_j, \quad F_{y,j} = u_j \sin \alpha_j, \quad (15)$$

turns the bilinear problem into a linear one. The decision vector becomes $z = [u_T, F_{x,1}, F_{y,1}, F_{x,2}, F_{y,2}]^\top \in \mathbb{R}^5$, and the wrench is produced by an extended configuration matrix built one column per decision variable,

$$B_e = [B_T \quad B_{F_{x,1}} \quad B_{F_{y,1}} \quad B_{F_{x,2}} \quad B_{F_{y,2}}], \quad (16)$$

$$B_{F_{x,j}} = \begin{bmatrix} 1 \\ 0 \\ -y_j \end{bmatrix}, \quad B_{F_{y,j}} = \begin{bmatrix} 0 \\ 1 \\ x_j \end{bmatrix}, \quad (17)$$

so that $\tau_c = B_e z$ is linear in z . Substituting the numerical thruster positions from Table II gives

$$B_e = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 12 & -3 & -13 & 3 & -13 \end{bmatrix}, \quad (18)$$

which has $\text{rank}(B_e) = 3$ (verified numerically), confirming the vessel can produce any demanded wrench $[F_x, F_y, M_z]^\top$ with this thruster layout. Once z is solved for, the physical commands are recovered as

$$u_j = \sqrt{F_{x,j}^2 + F_{y,j}^2}, \quad \alpha_j = \text{atan2}(F_{y,j}, F_{x,j}), \quad (19)$$

using atan2 rather than atan to resolve the correct quadrant unambiguously; the tunnel command is simply $u_T = z_1$ at its fixed angle α_0 .

D. Allocation Method: Minimum-Norm (Pseudo-Inverse) Solution

Because $B_e \in \mathbb{R}^{3 \times 5}$ is wide with full row rank 3, the equation $B_e z = \tau_c$ is underdetermined: infinitely many z satisfy it exactly for any achievable τ_c . Among all exact solutions, the implemented allocator selects the one of minimum Euclidean norm,

$$z^* = \arg \min_z \|z\|^2 \quad \text{s.t.} \quad B_e z = \tau_c = B_e^\top (B_e B_e^\top)^{-1} \tau_c, \quad (20)$$

the closed-form Moore–Penrose pseudo-inverse solution. Rather than computing $(B_e B_e^\top)^{-1}$ explicitly, the implementation solves the well-conditioned 3×3 linear system $B_e B_e^\top y = \tau_c$ for y (numerically preferred over an explicit matrix inverse) and recovers $z^* = B_e^\top y$. For the layout above,

$$B_e B_e^\top = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & -14 \\ 0 & -14 & 500 \end{bmatrix}, \quad \text{cond}(B_e B_e^\top) \approx 250, \quad (21)$$

which is well-conditioned, so the solve is numerically robust and the demanded wrench is reproduced exactly for any τ_c within the combined actuator envelope.

This minimum-norm choice is not arbitrary: since $F_{x,j}^2 + F_{y,j}^2 = u_j^2$ for each azimuth, the minimised cost $\|z\|^2 = u_T^2 + u_1^2 + u_2^2$ is exactly the sum of squared thrust magnitudes across all three thrusters – a standard convex proxy for propulsion power. The unweighted pseudo-inverse therefore already minimises a physically meaningful quantity; it does not, however, distinguish between thrusters of different capacity or account for actuator limits, as discussed next.

E. Limitations of the Pseudo-Inverse Allocator

The project text (§8.4) asks four questions of any allocation method, which are answered here honestly for the current implementation:

- **Are wrench components weighted equally?** Yes. The plain pseudo-inverse minimises an unweighted $\|z\|^2$, treating a newton of tunnel thrust, a newton of azimuth thrust, and (implicitly, through the shared moment arm terms) yaw authority as equally “expensive.” It does not account for the tunnel's much lower capacity ($u_{\max} = 32$ kN vs. 80 kN for the azimuths).
- **What happens if the requested wrench exceeds actuator capacity?** Nothing is enforced by the allocator: Eq. (20) is a closed-form solution with no awareness of $u_{\max}$, so it can in principle command $u_i > u_{\max,i}$. This is not merely theoretical: in Simulation 3 (Section VII-C) commanded thrust on Azimuth_2 reached approximately 198 kN, about $2.5 \times$ its 80 kN limit.
- **How are unnecessary azimuth rotations avoided?** They are not. The allocator accepts `u_now/alpha_now` but does not use them: each call solves Eq. (20) independently of the previous command, with no continuity or slew-rate penalty. This is consistent with Part 1, where thruster dynamics – including the azimuth-rate limit $\dot{\alpha}_{\max}$ of Table II – are disabled by default (Section V).

- **Is the problem linear or nonlinear?** Linear, by construction: the Cartesian substitution of Section VI-C removes the nonlinearity from the raw (u_j, α_j) formulation before the optimisation is solved.

The most visible consequence of both gaps together is discussed in Simulation 3 and 4, Section VII-C.

VII. SIMULATION RESULTS

All four mandatory simulations are run from `notebooks/part_1_demo.ipynb`, which produces the required time-history and xy -trajectory plots and checks each scenario against the automated pass/fail criteria.

A. Simulation 1: Station Keeping

Station keeping at $\eta_{sp} = [0, 0, 0]^T$ under (a) 0.5 m/s current from east and (b) 15 m/s mean wind from east.

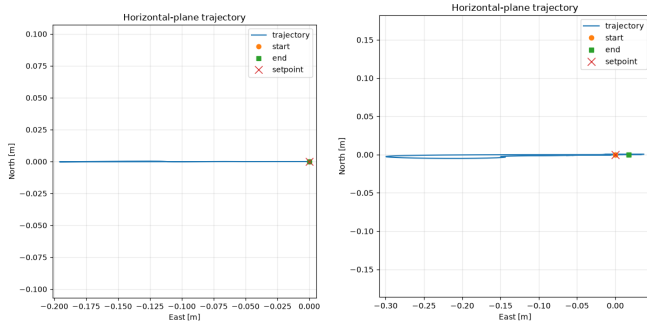


Fig. 1. Simulation 1 xy -trajectory: (a) current only (left), (b) wind only (right).

[TODO: Report max/steady-state position error, settling time, and actuator effort for each case using Figs. 1–4 above.]

B. Simulation 2: Rotating Current

Current direction rotates linearly from north to east over 300 s at a constant 0.5 m/s, no wind, vessel holding the origin.

[TODO: Discuss how the controller reacts during the 300 s rotation vs. behaviour once the current direction stabilises (see Fig. 5 for the current input and Fig. 6 for the resulting position hold / actuator effort).]

C. Simulation 3: Setpoint Change With and Without Reference Model

From $\eta_0 = [0, 0, 0]^T$ to $\eta_{sp} = [10, 10, 3\pi/2]^T$, no environmental forces, with and without the reference model. Since $3\pi/2$ is equivalent to $-\pi/2$, the controller correctly turns the vessel the short way: the logged heading converges to $\psi \approx -\pi/2$ rather than completing an unnecessary $+3\pi/2$ rotation.

Inspecting the per-thruster logs from this scenario (Fig. 9) reveals a notable allocator behaviour: both azimuth angles repeatedly swing by up to $\sim 160^\circ$ within a single 0.05 s time step (e.g. Azimuth_1: $-97.5^\circ \rightarrow 63.3^\circ$), with the two thrusters flipping together rather than independently. Cross-referencing against the commanded thrust magnitude at each jump shows $u_j \approx 0$ –1 N at exactly those instants – i.e. the

jumps occur when the demanded force from that thruster passes through zero. This is the angle-ambiguity effect noted in Section VI-E: Eq. (19) always returns a non-negative magnitude, so a continuous rotation through zero force is represented as a discontinuous $\sim 180^\circ$ jump to the opposite-signed solution rather than a smooth pass-through. The two azimuths flip together because of their mirrored position ($y = \pm 3$ m): a rotating demanded wrench crosses both thrusters' zero-force directions at close to the same time. The underlying demanded wrench itself also exceeds actuator capacity by $\sim 2.5\times$ during this scenario (Section VI-E), which is what repeatedly drives the allocation through these zero-crossings rather than passing through once.

[TODO: This oversized demand likely also reflects the controller not yet being tuned / lacking anti-windup (§6.3) – confirm with whoever owns the controller section, since it is the root cause of *why* the allocator is exercised this hard, even though the angle-jump behaviour itself is a structural property of the allocator and would appear (less dramatically) even with a well-tuned controller. Insert the actual position/heading/alpha(t)/u(t) plots here (with vs. without reference model) once finalised.]

D. Simulation 4: Four-Corner Test

Setpoints $\eta_0, \dots, \eta_5$ from Table 2 of the project text, 300 s hold per corner, no environmental forces.

[TODO: Verify and discuss whether the vessel reaches steady position before each transition (Fig. 10), and check whether the same azimuth-flip pathology from Simulation 3 (Fig. 9) reappears here at larger, sustained setpoint changes (Fig. 11).]

ACKNOWLEDGMENT

[TODO: Remove this section, or use it if relevant.]

REFERENCES

- [1] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. Chichester, UK: Wiley, 2011.
- [2] W. Blendermann, "Parameter identification of wind loads on ships," *Journal of Wind Engineering and Industrial Aerodynamics*, vol. 51, no. 3, pp. 339–351, 1994.

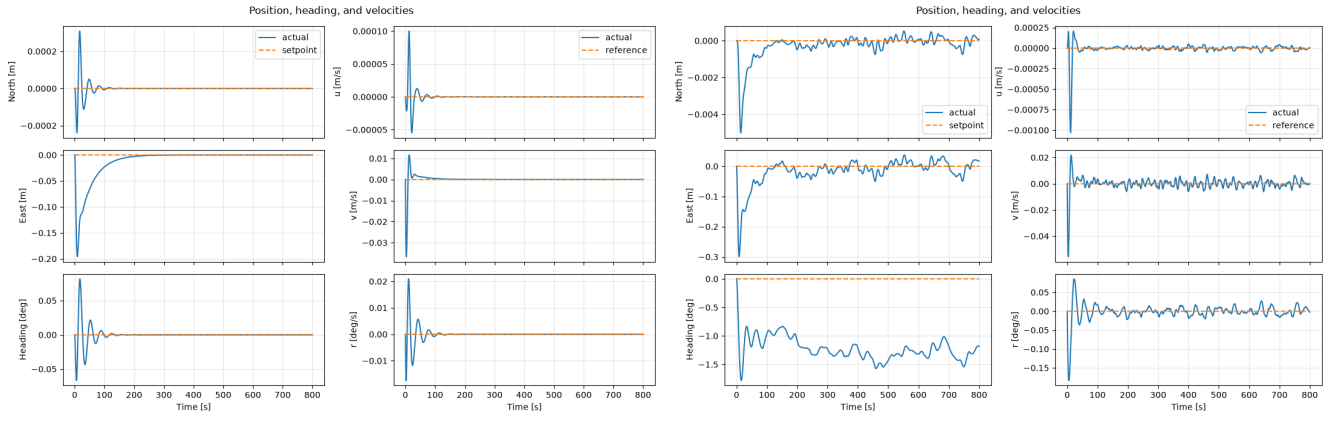


Fig. 2. Simulation 1 position, heading, and velocities vs. setpoint: (a) current only (left), (b) wind only (right).

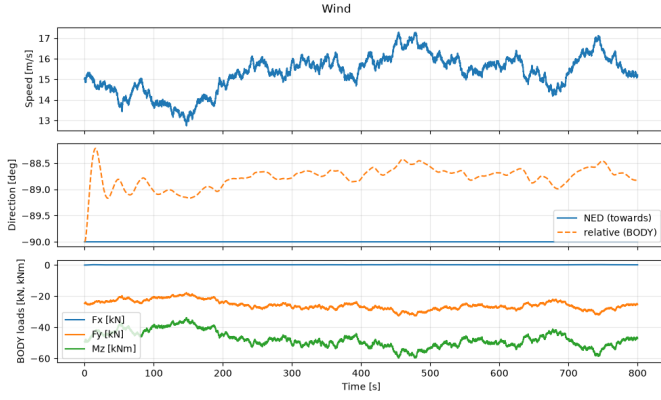


Fig. 3. Simulation 1b: relative wind speed, direction, and body-frame components driving the wind load.

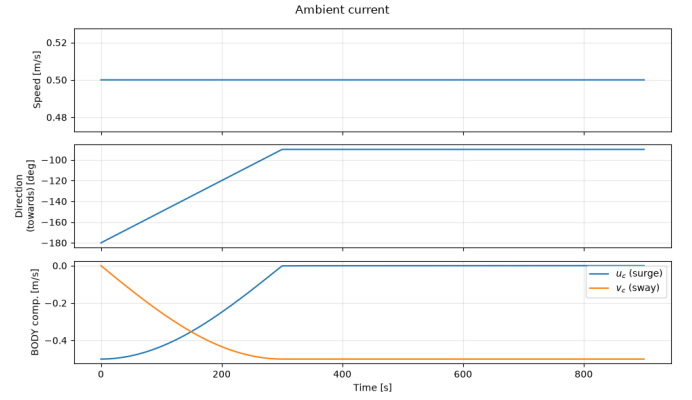


Fig. 5. Simulation 2: ambient current speed, direction, and body-frame components during the 300 s rotation.

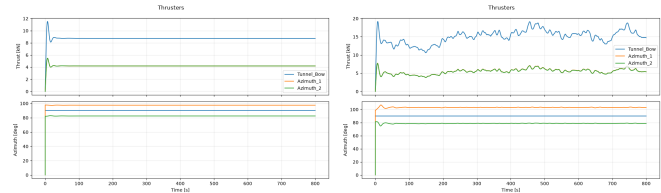


Fig. 4. Simulation 1 thruster effort: (a) current only (left), (b) wind only (right). Both stay well within $u_{\max}$.

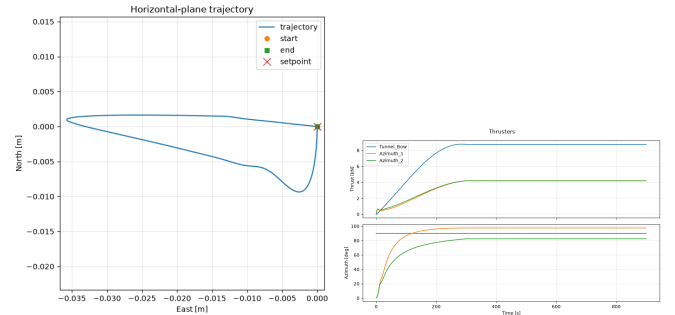


Fig. 6. Simulation 2: xy -trajectory (left) and thruster commands (right) during the current rotation.

Simulation 3: setpoint step with / without reference model

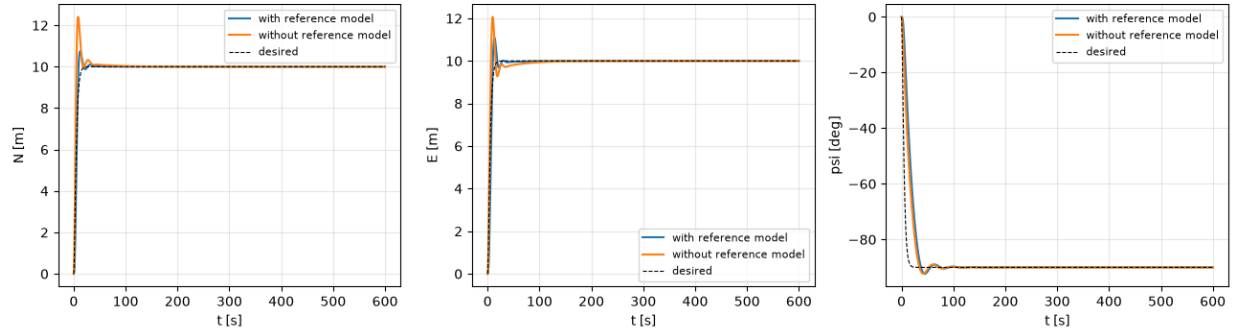


Fig. 7. Simulation 3: N , E , and ψ with vs. without the reference model, against the desired trajectory. Both runs converge cleanly to the setpoint – the pathology below is invisible at this level.

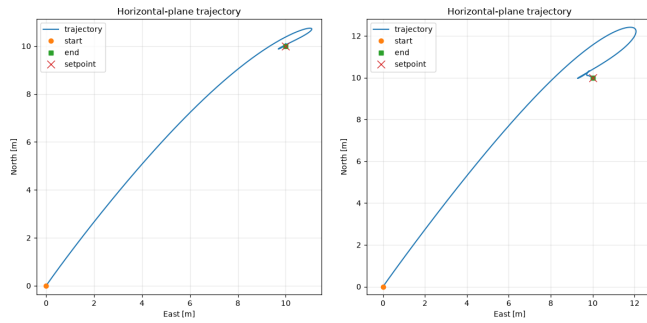


Fig. 8. Simulation 3 xy -trajectory: with reference model (left) vs. without (right).

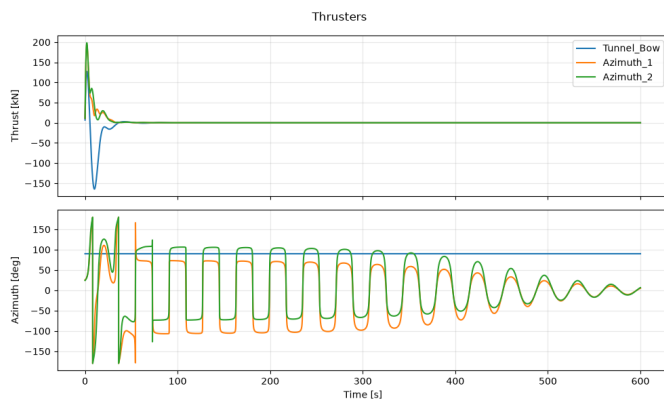


Fig. 9. Simulation 3 (with reference model) thruster commands: an initial spike to ~ 200 kN (vs. an 80 kN limit), followed by a slowly-damping azimuth oscillation for the entire 600 s run – invisible in Fig. 7, but not in the underlying actuator commands.

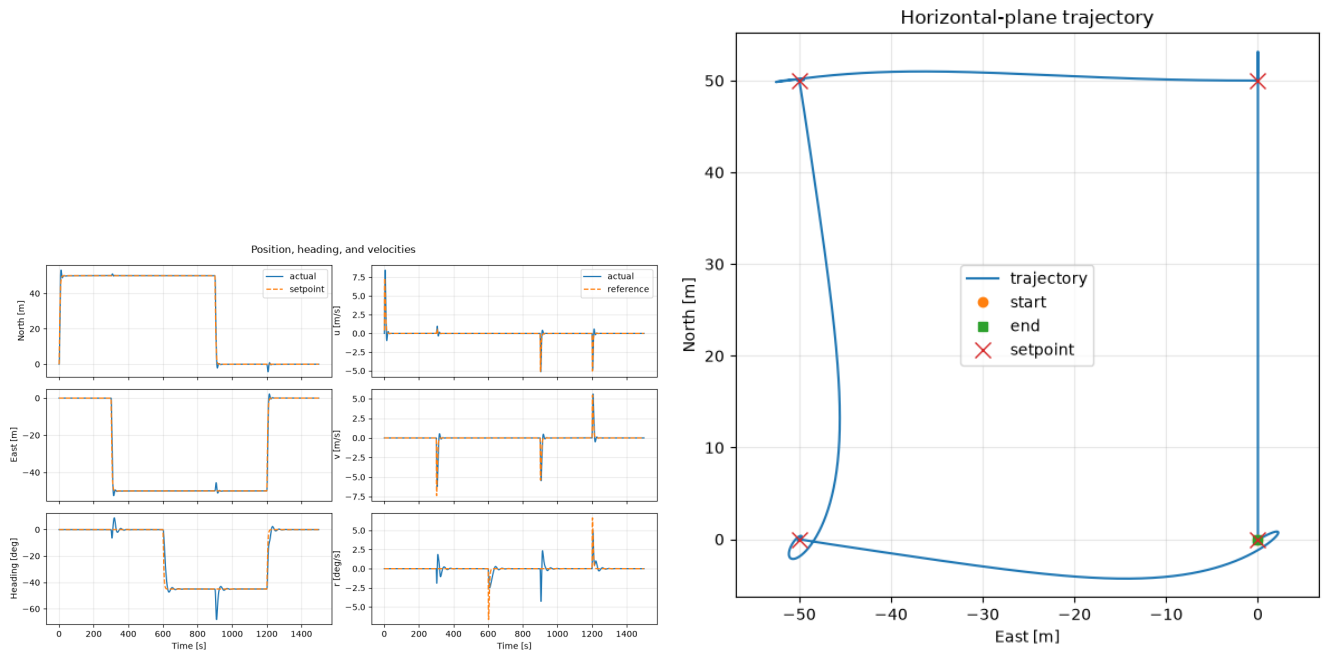


Fig. 10. Simulation 4 (four-corner test): position, heading, and velocities vs. setpoint (left); xy -trajectory through all five corners (right).

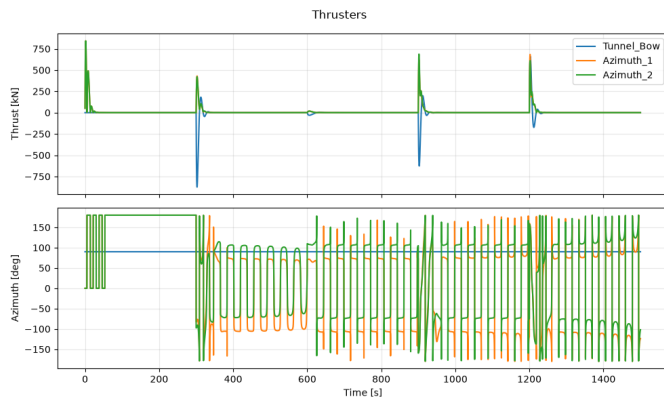


Fig. 11. Simulation 4 thruster commands across all five corners.